SET A

# BRAC UNIVERSITY
## Department of Computer Science and Engineering

Examination: Semester Midterm | Semester: Fall 2023
Duration: 1 Hour 15 Minutes | Full Marks: 40

## CSE 422: Artificial Intelligence

Answer the following questions.
Figures in the right margin indicate marks.

**1. CO1** Suppose you are asked to generate a set of 5 real numbers in the range of -10 to 10 whose mean would be 2. Now, you have to solve this problem using Genetic Algorithm.

a. **Encode** the problem and deduce two parent chromosomes, PC1 and PC2. 3

b. **Define** a suitable fitness function for the problem and calculate the fitness of PC1 and PC2. 2

c. **Illustrate** a single-point crossover at any point you prefer between PC1 and PC2, then perform mutation. You can mutate only two numbers of your choosing. 2

d. **Explain** briefly the role of crossover in finding the optimal solution in such types of problems. 3

**2. CO1**

a. **Determine** which of the following statements are true or false. Give reasons behind your answers. 7.5

-A simple reflex agent is suitable to work in a partially observable environment
-A tic-tac-toe game is an example of a stochastic environment
-Hill Climbing allows backtracking
-For simulated annealing, the minimum value for $e^{\Delta E/T}$ is negative infinity
-Hill Climbing is guaranteed to find the global optima

b. **Define** what a local maxima is concerning the hill climbing algorithm. Then using a state space graph present a scenario that depicts a local maxima for the 8 puzzle problem. [Do not need to formulate the entire state space. Just show a subset of the state space graoh] 2.5

**3. CO1**

a. Minimax search is breadth-first: it processes all the nodes at a level before moving to a node in the next level. Is the statement True/False? **State** your reason 2

b. **State** whether it is possible that for a given search tree Minimax and Alpha-Beta Pruning will end up visiting an equal number of terminal nodes. [Yes/No] 1

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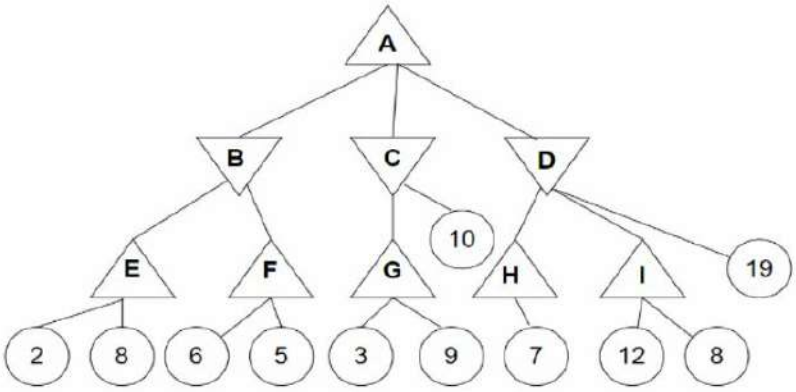
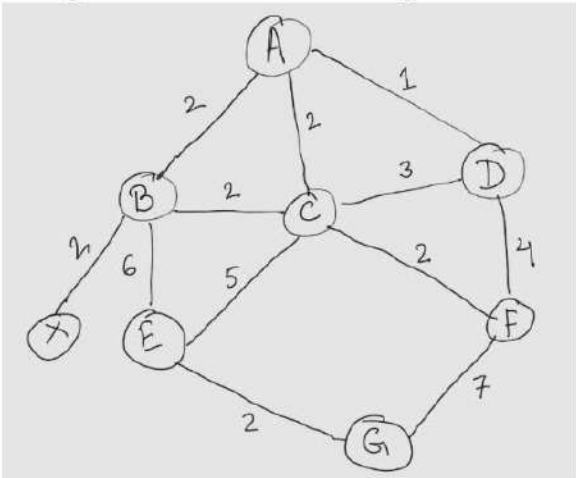
c. Given the following snippet of state space tree of a two player game, **apply** alpha-beta pruning showing alpha,beta and node values for all the nodes. 7




**4.CO1** In the graph below let, A, B, C, D, E, F, G, X denote different states of a problem space, where the edge between the states describes the action, and the number along the edge is the cost of the action to go from one state to another.




Let, A be your initial state and G be your goal. In the table below we give you the heuristic value to reach G from any given node, n in the graph.

| n | A | B | C | D | E | F | X | G |
|---|---|---|---|---|---|---|---|---|
| h(n) | 9 | 6 | 7 | 7 | 2 | 5 | 1 | 0 |

a. From looking at the graph, **find** the optimal path from A to G (DO NOT USE ANY SIMULATION), and show that the given heuristic for A is valid. 1

b. Using the given graph and the given values, **explain** why the Greedy best search does not guarantee an optimal solution. 4

c. For A* algorithm to be optimal, which condition on the heuristics should be imposed on the above graph. **Explain**. 1

d. **Check** if the heuristics for A, C and D satisfy the optimality condition of A* algorithm. Change the values of the heuristics for violating state(s), so that the condition is satisfied. 4

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Duration: 1 Hour 30 Minutes

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Full Marks: 40

CSE 422: Artificial Intelligence

Answer all the following questions.

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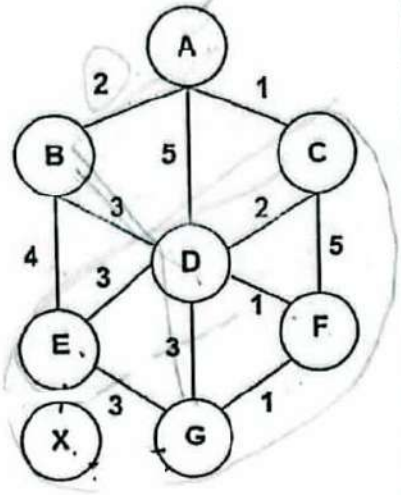
| Name: | ID: | Section: 01 |
|---|---|---|

**1. CO1**




| States,n | h1(n) | h2(n) |
|---|---|---|
| A | 5 | 5 |
| B | 5 | 4 |
| C | 3 | 2 |
| D | 2 | 2 |
| E | 3 | 2 |
| F | 1 | 1 |
| X | ?? | ?? |
| G | 0 | 0 |

Consider in the above graph, A is the initial state.

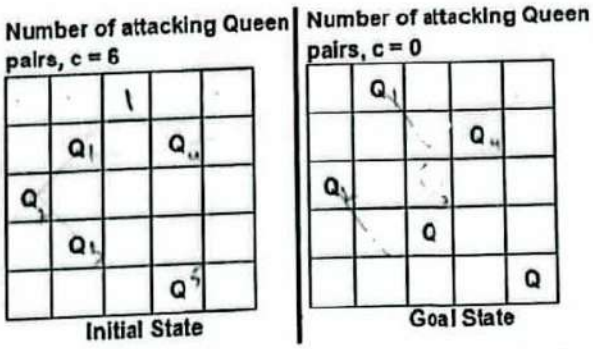
a. **Examine** node "X" and assign it a heuristic value. Provide reasons behind your logic. 2

b. Assuming both h1(n) and h2(n) are admissible, **apply** custom graph A* search on the given graph using the better heuristic. In the search, you replace an already existing node in the fringe only if the same node is visited again at a lower f-cost. 4

c. Suppose C is the initial state and E is the goal state. **Determine** whether an A* search using either h1(n) or h2(n) will return the optimal path from C to E. 2

d. If the cost from B to E and B to D becomes 3 and 2 respectively, then **determine** whether h1(B) 's heuristic value remains consistent. Show calculations 2

**2. CO1**




a. **Observe** the given states of a 5-Queen Problem. Now, sketch a portion of the state space tree (branching factor = 2) and showcase a scenario where hill climbing would get stuck in a Local Maxima. At each depth, you can select only one Queen and move it one step Up, Down, Left, or Right. [Note: You don't have to show the complete state space tree] 5

b. **Determine** how you can solve the problems of Hill Climbing Search. 3

c. **Examine** how Local Search is different from Informed Search. 2

Set A

3. CO1 Ruhan is a salesperson at B.Daily products. He is given a list of 5 cities with their corresponding coordinates on a 2D plane. The cities are numbered from 1 to 5 where 5 is the total number of cities. His objective is to find the most optimal route that visits all 5 cities exactly once and returns to the starting point. The optimality is measured based on the lowest distance traveled. The distance between two cities is calculated using the Euclidean formula. Euclidean Distance Formula points (x1, y1) and (x2, y2) is as follows:

$$\text{Euclidean_distance} = \sqrt{(x1 - x2)^2 + (y1 - y2)^2}$$

| City 1 | City 2 | City 3 | City 4 | City 5 |
|---|---|---|---|---|
| (2,3) | (5,1) | (1,4) | (4,2) | (3,5) |

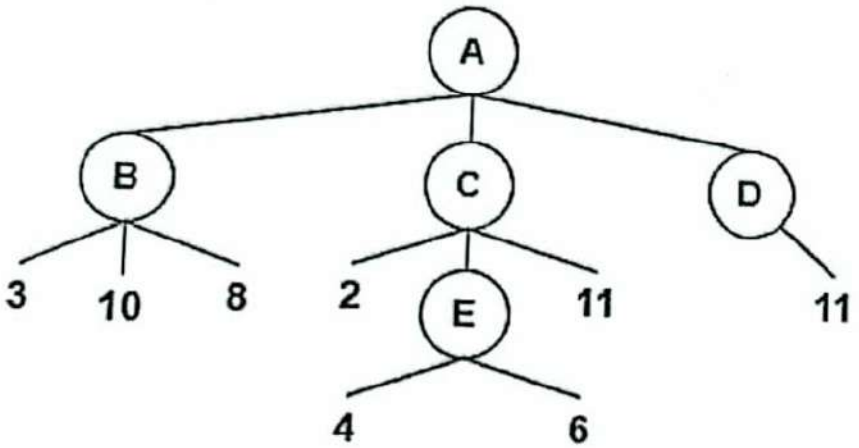
Now you are well-aware of an optimization algorithm named Genetic Algorithm. Your task is to help Ruhan achieve his goal using the following steps:

a. Encode the problem and create four parent chromosomes. Then **determine** an appropriate fitness function and choose the two fittest chromosomes. 4

b. **Apply** two-point crossover on the chromosomes derived from (a) to produce two offspring. [NB: The two-point crossover technique chooses two random points from chromosomes and genes are exchanged at these points]. 3

c. **Examine** how you can mutate the offspring derived from step (b) and comment on the fitness of the final produced offspring. 3

4. CO1




a. **Apply** alpha-beta pruning on the given game tree to find the best choice when at state A. Showcase the values of the alpha, beta, and the final value of the nodes. Also, show the pruned branches. 8

b. **Examine** how the parameters alpha and beta in the alpha-beta pruning algorithm help to reduce the search space. 2

Set B

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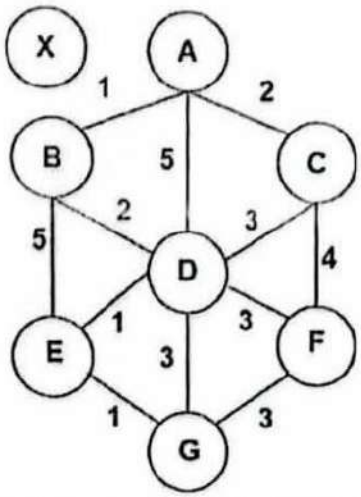


| States,n | h1(n) | h2(n) |
|---|---|---|
| A | 5 | 5 |
| B | 2 | 3 |
| C | 4 | 5 |
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| G | 0 | 0 |

Consider in the above graph, A is the initial state.

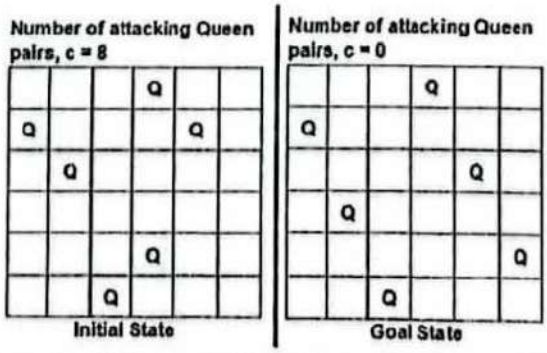
a. **Examine** node "X" and assign it a heuristic value. Provide reasons behind your logic. 2

b. Assuming both h1(n) and h2(n) are admissible, **apply** custom graph A* search on the given graph using the better heuristic. In the search, you replace an already existing node in the fringe only if the same node is visited again at a lower f-cost. 4

c. Suppose C is the initial state and E is the goal state. **Determine** whether an A* search using either h1(n) or h2(n) will return the optimal path from C to E. 2

d. If the cost from C to F and C to D becomes 3 and 2 respectively, then **determine** whether h2(C) 's heuristic value remains consistent. Show calculations 2

**2. CO1**




a. **Observe** the given states of a 6-Queen Problem. Now, sketch a portion of the state space tree (branching factor = 2) and showcase a scenario where hill climbing would get stuck in a Local Maxima. At each depth, you can select only one Queen and move it one step Up, Down, Left, or Right. [Note: You don't have to show the complete state space tree] 5

b. **Explain** Plateau in a Hill Climbing Search problem. 3

c. **Explain** how Simulated Annealing is different from Basic Hill Climbing Search. 2

Set B

**3. CO1** Ruhan is a salesperson at B.Daily products. He is given a list of 5 cities with their corresponding coordinates on a 2D plane. The cities are numbered from 1 to 5 where 5 is the total number of cities. His objective is to find the most optimal route that visits all 5 cities exactly once and returns to the starting point. The optimality is measured based on the lowest distance traveled. The distance between two cities is calculated using the Euclidean formula. Euclidean Distance Formula points (x1, y1) and (x2, y2) is as follows:

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| City 1 | City 2 | City 3 | City 4 | City 5 |
|---|---|---|---|---|
| (8,2) | (4,6) | (1,3) | (7,5) | (5,1) |

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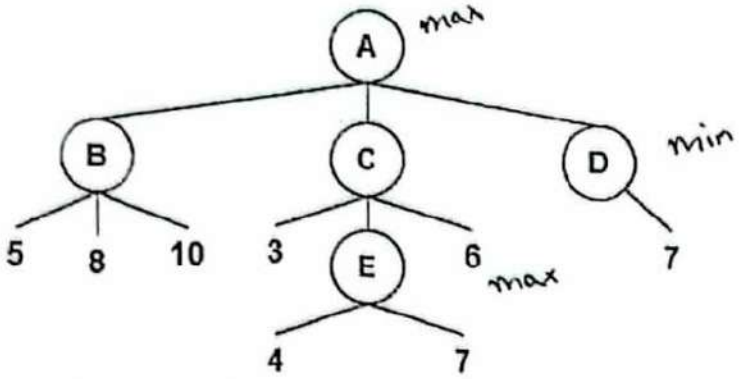


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